

Estimation of Cluster-Specific Causal Effects on Spatially Associated Survival Data Using SoftBART

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The Problem: Treatment Delay in Breast Cancer

Does delaying breast cancer treatment (>90 days) cause worse survival? And how does the effect vary across geographic regions?

Breast Cancer Facts

Most diagnosed cancer in US women

2nd leading cause of cancer death

Treatment Delay (TD): diagnosis → first treatment

TD > 90 days ⇒ worse 5-yr survival

Florida Cancer Registry (FCR)

$N = 68,009$ patients, $K = 67$ counties

41.3% right-censored

$Z = 1$ (TD ≤ 90 d) vs. $Z = 0$ (TD > 90 d)

Confounders: Age, HR status, Tumor Grade, Biopsy Delay

Key Challenges

Data

- **Spatial dependence** — neighboring counties share profiles
- **Right-censoring** (41.3%)
- **Confounded** treatment (not random)
- **Clustered** — patients within counties

Modeling

- **Non-linear** treatment–confounder–outcome links
- **Unmeasured** county-level confounding
- Need **county-** & **patient-specific** effects
- No method does spatial + nonparametric + survival

Goal & Causal Estimands

Counterfactual survival: $S_{ij}^{(z)}(t \mid \mathbf{x}_{ij}) = P(T_{ij}(z) \geq t \mid \mathbf{x}_{ij}, \text{County} = i)$

Patient-level — CERM (restricted mean gain over $(0, t^*)$):

$$\text{CERM}_{ij}(t^*) = \mathbb{E}[\{T_{ij}(1) \wedge t^*\} - \{T_{ij}(0) \wedge t^*\} \mid \mathbf{x}_{ij}, i]$$

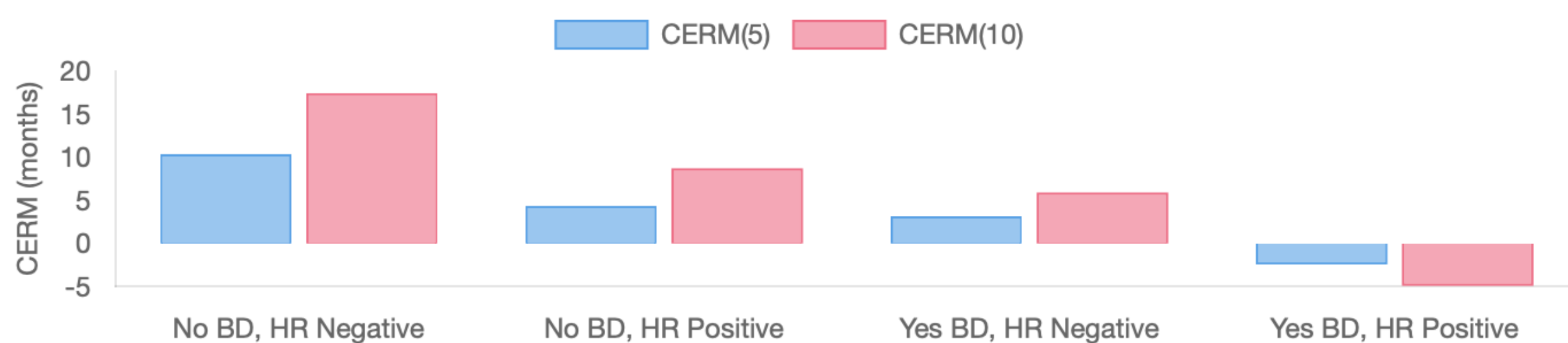
Multi-resolution aggregation:

County: $\text{ACERM}_i(t^*) = \frac{1}{n_i} \sum_j \text{CERM}_{ij}(t^*)$ (also for treated/untreated: ACERMT, ACERMU)

State: $\text{RATE}(t^*) = \frac{1}{N} \sum_{i,j} \text{CERM}_{ij}(t^*)$ (RATT, RATU by group)

Survival-prob effect: $\text{SPTE}_i(t) = S_i^{(1)}(t) - S_i^{(0)}(t)$

Identifiability under: SUTVA, Consistency, Unconfoundedness, Positivity, Non-informative censoring.



Heterogeneous CERM (months) by Biopsy Delay & HR status — effects differ sharply across subgroups, motivating a *flexible* model.

Proposed Method: PS-LND (Soft BART + DAGAR)

Data $(y_{ij}, \delta_{ij}, z_{ij}, \mathbf{x}_{ij}) \Rightarrow$ **Stage 1:** estimate PS $\hat{e}_{ij} \Rightarrow$ **Stage 2:** survival outcome $\Rightarrow$ **causal effects**

Stage 1 — Propensity Score (Spatial Probit-SBART)

$$P(Z_{ij} = 1 \mid \mathbf{x}_{ij}, V_i) = \Phi(\underbrace{b(\mathbf{x}_{ij})}_{\text{SBART}} + \underbrace{V_i}_{\text{DAGAR}}), \quad \hat{e}_{ij} = E[\Phi\{b(\mathbf{x}_{ij}) + V_i\} \mid \mathcal{D}_1]$$

$b(\cdot)$: unknown function with Soft BART prior; $\mathbf{V} \sim N(\mathbf{0}, \tau_v^{-1} \mathbf{Q}(\rho_v)^{-1})$ captures **unmeasured spatial confounding**

Stage 2 — Outcome (Log-Normal AFT with BCF structure)

$$\log T_{ij} \sim N\left(\underbrace{\mu(\mathbf{x}_{ij}, \hat{e}_{ij})}_{\text{prognostic SBART}} + \underbrace{\tau(\mathbf{x}_{ij})}_{\text{treatment SBART}} \cdot Z_{ij} + \underbrace{W_i}_{\text{county DAGAR}}, \sigma^2\right)$$

Separate prognostic μ & treatment τ surfaces handle **targeted selection**; $\hat{e}_{ij}$ enters as a confounder

$\mathbf{W} \sim N(\mathbf{0}, \tau_w^{-1} \mathbf{Q}(\rho_w)^{-1})$: county random effects; right-censoring via data augmentation

DAGAR spatial prior — interpretable smoothing. For county i with preceding neighbors $\mathcal{N}(i)$:

$$W_i \mid W_{\mathcal{N}(i)} \sim N\left(\frac{\rho_w}{1+(n_i-1)\rho_w^2} \sum_{j \in \mathcal{N}(i)} W_j, \frac{1+(n_i-1)\rho_w^2}{\tau_w(1-\rho_w^2)}\right)$$

$\rho_w \rightarrow 0$: counties independent; $\rho_w \rightarrow 1$: strong neighbor smoothing

Pairwise correlations stay $\approx \rho_w$ — genuine “borrowing of strength”

Why PS-LND Wins — Key Novelties

- ✓ **Flexible & nonparametric:** Soft BART’s *smooth* sigmoid trees ✓ fit nonlinear effects with *no* pre-specified interactions
- ✓ **Interpretable spatial (DAGAR):** proper, sparse, positive- ✓ definite precision; $\rho_w \in (0, 1)$; scalable
- ✓ **Doubly robust:** consistent if *either* PS or outcome model is correct ✓
- ✓ **No feedback:** two-stage design isolates PS from a misspecified outcome; efficient
- ✓ **Targeted selection (BCF):** splits prognostic vs. treatment effect for clean heterogeneity
- ✓ **County-specific & multi-level:** full posterior for patient, county & state effects

Simulation: Verifying the Method

$K=35$ clusters on a 5×7 grid, $N=1750$, 5 confounders, 150 replications. Method stress-tested under three regimes:

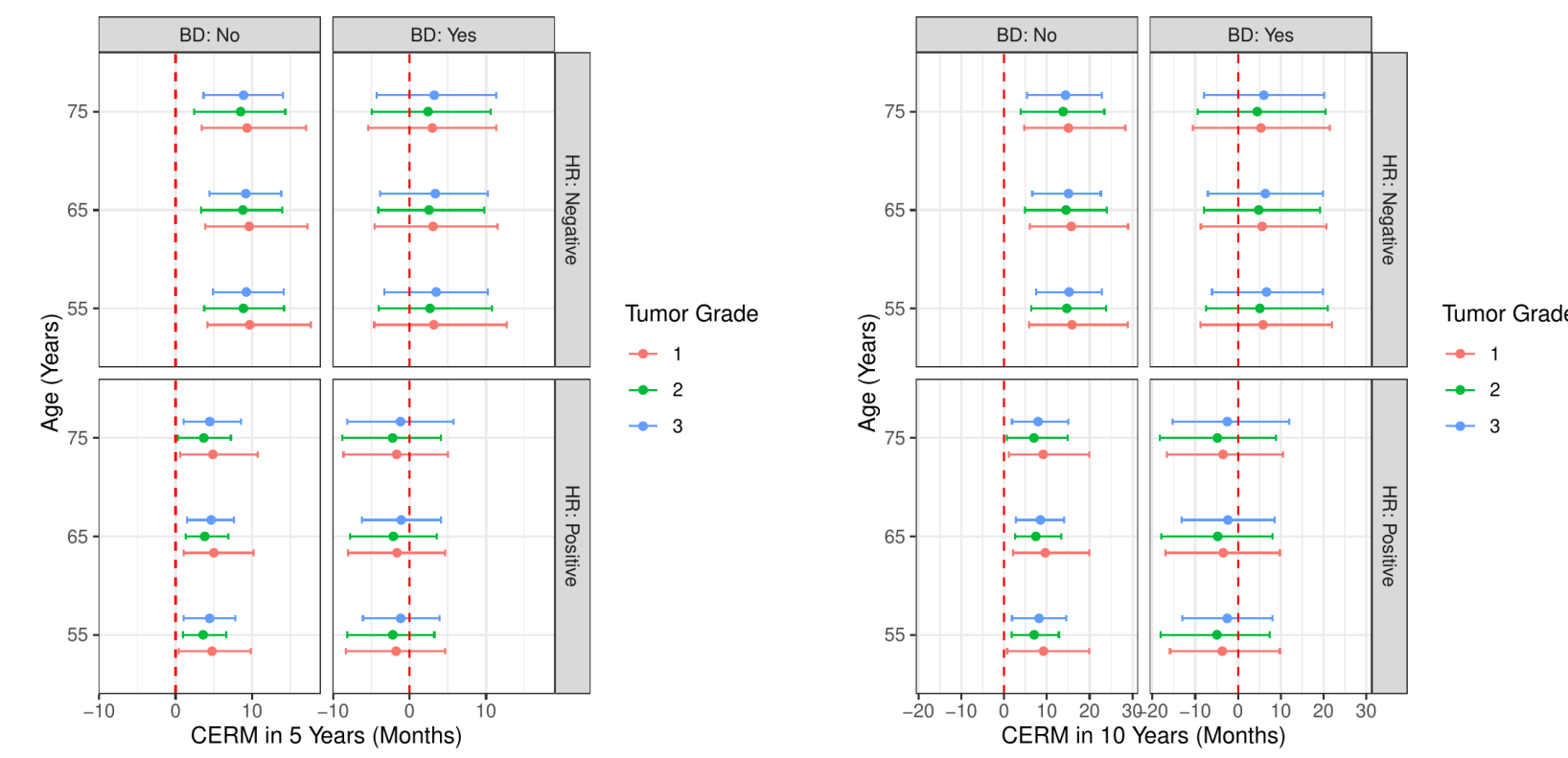
Scenario	What it tests	Data-generating truth
1. Ideal	Correct specification	Log-Normal + DAGAR effects ($\rho_w = 0.7$); nonlinear main effects + treatment interactions
2. Misspecified	Robustness to model error	Adds covariate $\times$ random-effect interactions $g(\mathbf{x}, z, w)$ omitted by fitted model
3. Stress	Distribution violation + heavy censoring	Weibull (not log-Normal), 50% censoring, <i>strong</i> confounding test

Scenario	Metric	PS-LND	riAFTBART	PSW-Fr.	IPW
1. Ideal	Cov / MAE	0.93–0.97 / 0.01	0.46–0.95 / 0.08–0.58	0.54–0.97 / 0.08–0.50	0.38–0.52 / 0.06
2. Misspec.	Cov / MAE	0.92–0.95 / 0.01	0.77–0.89 / 0.20–0.79	0.64–0.89 / 0.21–0.63	0.42–0.53 / 0.06
3. Stress	Cov / MAE	0.87–0.90 / 0.03	0.55–0.68 / 0.24–0.66	0.76–0.82 / 0.08–0.21	0.42–0.47 / 0.53

PS-LND attains near-nominal coverage & the lowest MAE in every regime — the only method robust to misspecification *and* distributional violation.

Application: Florida Cancer Registry — WA-Distant Stratum

Patient-Level Effects (Broward County)



(a) 5-year CERM estimates.

(b) 10-year CERM estimates.

Fig 1: Point and 95% interval estimates of CERM (in months) restricted within intervals of (a) 5 and (b) 10 years for WA patients with Distant-stage cancer in Broward County

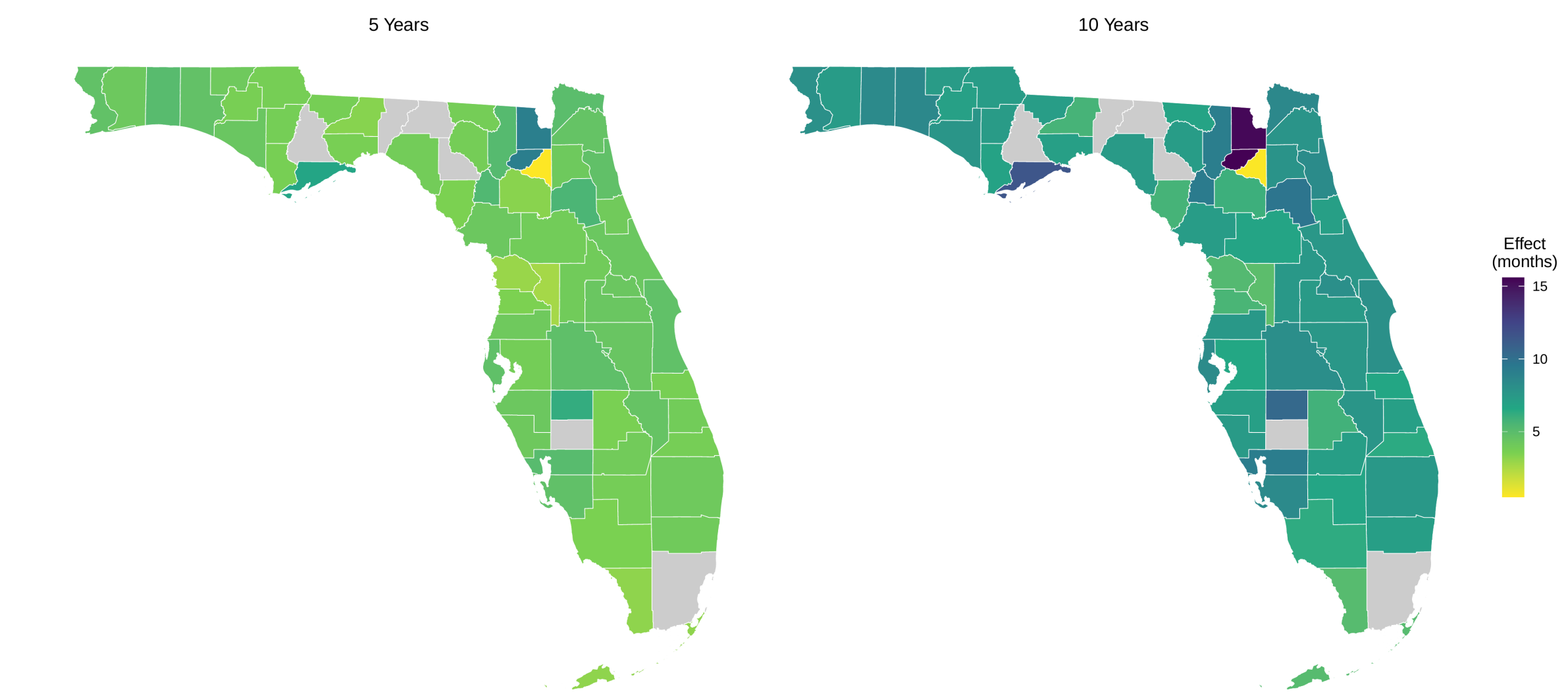
$\widehat{\text{CERM}}$ (months) at 5 & 10 yr by Age, BD, HR, Tumor Grade ($n=183$)

HR–, no BD: **13.8–15.9 mo.** gain (10 yr)

HR+, no BD: **6.9–9.6 mo.** gain

With biopsy delay: no significant effect (CI $\ni 0$)

County-Level Spatial Patterns



$\widehat{\text{ACERM}}$ at 5 (left) & 10 (right) yr. Yellow = 0 mo → dark = 15+ mo; gray = no data

Highest: Union (>15 mo.), Baker (>13 mo.)

Near-zero: Bradford (CI $\ni 0$)

Clear **spatial clustering** of treatment effects

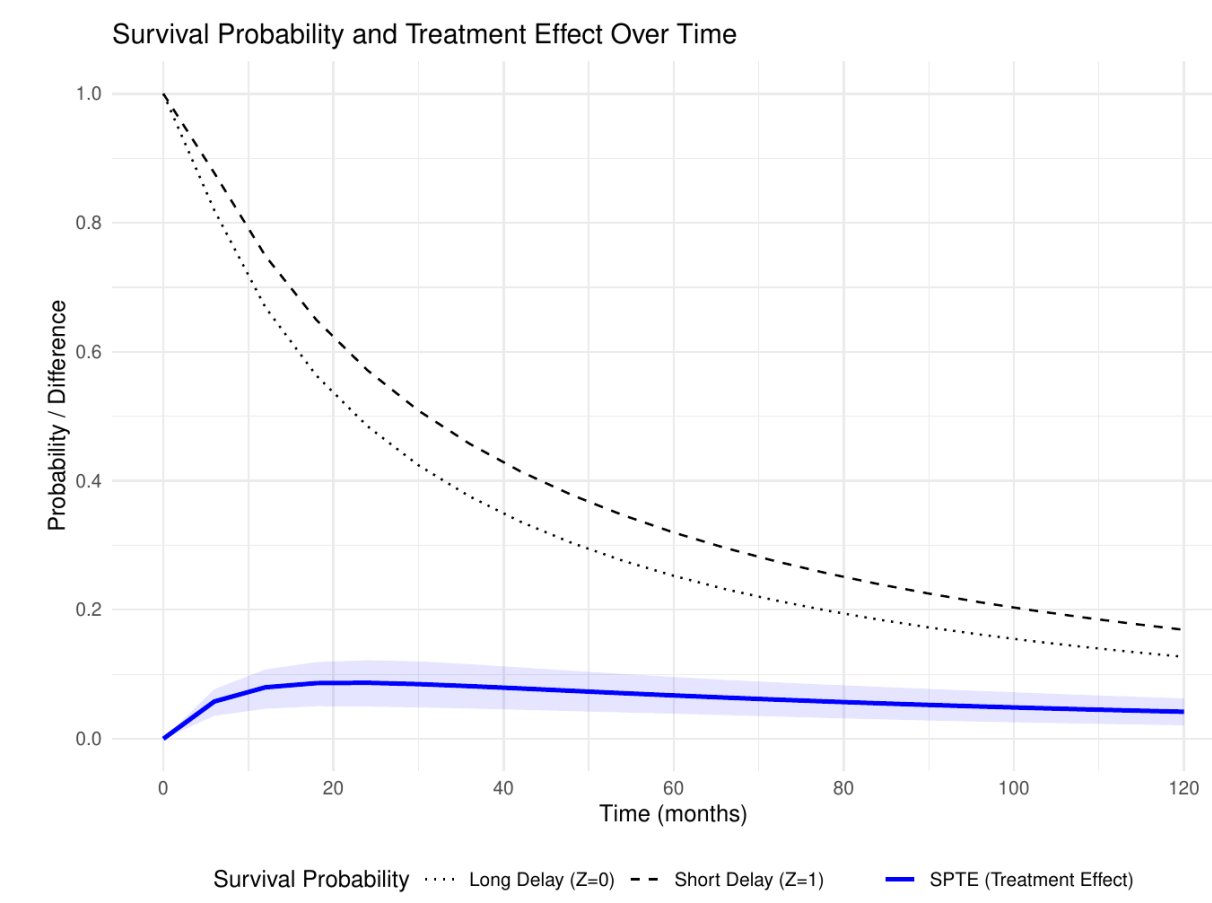
State-Wide Treatment Effects

Estimand	N	Est. (mo.)	95% CI
RATE (10 yr)	1994	7.62	(4.42, 10.86)
RATT (10 yr)	687	6.74	(3.09, 10.08)
RATU (10 yr)	1307	8.09	(4.60, 11.42)
RATE (5 yr)	1994	4.43	(2.56, 6.23)

No CI contains zero $\Rightarrow$ strong evidence

RATU > RATT: long-TD patients gain the most

Benefits persist beyond 5 years



$\widehat{\text{SPTE}}(t)$: peak 11% survival-probability gain at $t=2.2$ yr; CI > 0 throughout

Policy Takeaway

Short TD $\Rightarrow$ **7.6 mo.** average survival gain statewide

Priority: **HR– patients without biopsy delay**

Target **rural northern counties** for patient-navigation programs